Full Pair-Entropy Study from Existing MD Trajectories

Protocol for the eight-mixture charged-fluid study

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Abstract

This protocol plans an independent MD determination of the complete all-species pair entropy s_2, selected probe-centered contributions, and their relationship to thermodynamic-integration (TI) entropy and self-diffusion. The primary study uses the existing eight compositions, three replicas each, at $T^* = 2$ and $\rho^* = 0.7$. All-particle positions permit extraction of every partial radial distribution function (RDF) without new MD. The main challenges are nonlinear histogram bias, rare ion-pair statistics, finite-size normalization, and radial convergence. The protocol defines checks and decision criteria before fitting entropy–diffusion relations. No new production simulations are submitted by this planning document.

1 Questions, scope, and existing evidence

The primary question is whether replacing total excess entropy by full pair entropy preserves the conclusions of our replicated MD study. Secondary questions are whether the remainder $s_{\mathrm{ex}}^{\mathrm{TI}} - s_2^{\mathrm{MD}}$ changes with physical solvent diversification, and whether ion-centered structural contributions explain probe mobility better than the solvent-dominated bulk sum. Rosenfeld-type scaling [1] and its mixture/tracer extensions [2] motivate these tests, but no universal relation is assumed. The local pair statistic described by He et al. [3] motivates a separate selected-pair comparison; it is not interchangeable with the full mixture entropy.

Use all eight states B3, B5, U4, L5, M5, H5, U6, and H7, with three replicas per state. All contain 120 particles of each divalent sign, out of $N = 4000$. Exact compositions and type definitions are in `ComponentEntropyStudy_20261030/selected.json`; the composition tables are reproduced in the existing reports. The primary contrasts are L5–U4 (label null), M5–U4 and H5–U4 (moderate and strong solvent diversification), H7–U6 (diversification with monovalent background), and B5–B3 (salt addition). Broad baseline-to-design comparisons are secondary.

Existing HNC outputs preserve the same qualitative ordering for full pair and total entropy. For H5–U4 and H7–U6, HNC pair increments are about $0.075k_B$ versus total HNC increments near $0.111k_B$. These preliminary numbers are recorded in `doc/hnc_pair_entropy_comparison.json`. They are hypotheses for an MD test, not MD pair-entropy measurements. The completed diffusion estimates, including the provisional approximately 5% controlled probe response, remain unchanged by this analysis.

2 Input data and feasibility

Each dense production trajectory is located under the study directory at

`md/<case>/replica_<r>/transport/positions.nc`.

The files contain all-particle `unwrapped_coordinates`, IDs, types, frame times, cell origins, lengths, and angles. A header inventory is stored in `doc/md_pair_entropy_trajectory_inventory.json`. The standard record has 2001 frames, $N = 4000$, and positions every 0.5τ over 1000τ. The cubic

box length is approximately 17.878σ, giving a maximum complete spherical radius of approximately 8.939σ.

Use these position records, not the velocity files: the latter contain only subsets of neutral species. Resolve types and charges against the immutable input `transport/data.mixture`, `selected.json`, and replica manifest. Check frame IDs and sorting explicitly. Read time scale factors and box metadata through the NetCDF interface rather than assuming raw array values are already physical times.

Primary structural averages use the final 500τ of dense production, matching the TI averaging interval at the dense endpoint. Analyze the full 1000τ and its two halves as stationarity/sampling sensitivities. First benchmark 20 frames from one replica of B3 and H7; then process all replicas with matched frame counts. Increase from 50 to 100, 200, 500, and, where needed, all approximately 1001 frames in the final half. All particles are retained; only frame thinning is allowed in the initial sampling pilot.

A direct $N = 4000$ calculation has 7,998,000 unordered pairs per frame. Full-frame analysis across 24 replicas therefore requires substantial postprocessing despite requiring no new dynamics. Measure time, memory, and I/O in the pilot before selecting CPU allocation or an offline GPU implementation. Stream coordinates and accumulate histograms in a compiled pair-distance kernel; avoid storing all distances or loading every trajectory at once. Parallelize independent replicas within measured resource limits. Do not rerun forces or electrostatics merely to obtain distance histograms.

3 RDF estimation and finite-system conventions

Wrap positions into the periodic box and calculate minimum-image separations. This is a static observable, so no COM subtraction is needed. Only the subsequent transport comparison uses the established common-system COM frame. Exclude self-pairs and count each physical pair once.

For a bin $[r_k, r_{k+1})$, use its exact spherical shell volume

$$\Delta V_k = \frac{4\pi}{3}(r_{k+1}^3 - r_k^3). \tag{1}$$

If $H_{ab,k}$ is the unordered histogram accumulated over F frames of a fixed-volume system, define the ideal-normalized finite-system RDF as

$$g_{ab,k} = \frac{H_{ab,k}}{F M_{ab} \Delta V_k / V}, \qquad M_{ab} = \begin{cases} N_a N_b, & a \neq b, \\ N_a(N_a - 1)/2, & a = b. \end{cases} \tag{2}$$

This convention gives unity in expectation for ideal random positions at fixed species counts. Document any conversion to the alternative bulk-density normalization, especially for small ionic populations. Do not let a diagonal $1 - 1/N_a$ offset accumulate as an apparent long-range entropy contribution. If the box varies, accumulate the frame-dependent ideal counts explicitly; our intended records are constant-volume and must be checked as such.

Compute bins from zero to a common radius below the smallest half-box length. Use $\Delta r = 0.02\sigma$ initially, with 0.01, 0.04, and where useful 0.08σ as resolution checks. Do not extend the spherical normalization beyond half the box. Record both raw counts and normalized RDFs, including metadata and counts for every species pair. Do not average RDFs from different replicas before estimating replica uncertainty.

Mechanically identical secondary solvents in L5 must be merged for the primary physical-species calculation. Independently process an explicitly label-resolved version as a diagnostic. Nonlinear entropy estimates from noisy, subdivided histograms need not be exactly label invariant even when interactions are; merging the counts before estimating the physical RDF avoids that artifact. Check actual U4/L5 positional identity by ID, and also perform an exact synthetic split/merge on the same snapshots.

4 Entropy integrals and pair decomposition

For $I_{ab}(R) = \int_0^R \phi(g_{ab}(r))r^2\,dr$, where $\phi(g) = g\ln g - g + 1$, calculate the full pair entropy

$$\frac{s_2(R)}{k_B} = -2\pi\rho \sum_{ab} x_a x_b I_{ab}(R). \tag{3}$$

The sum is over ordered species pairs. With an unordered pair list, store

$$C_{aa} = -2\pi\rho x_a^2 I_{aa}, \qquad C_{ab} = -4\pi\rho x_a x_b I_{ab} \quad (a < b), \tag{4}$$

so that $s_2/k_B = \sum_{a\le b} C_{ab}$. This makes the cross-pair factor of two explicit. For histogram RDFs, integrate using exact radial bin weights $(r_{k+1}^3 - r_k^3)/3$ rather than an inconsistent midpoint volume. At $g = 0$, use $g\ln g = 0$ and $\phi(0) = 1$; do not discard empty core bins or apply an arbitrary density floor.

Store full cumulative integrals and decompose them into neutral–neutral, neutral–ion, ion–ion, divalent-containing, and monovalent-containing pairs. Categories must be defined without overlap when their sum is compared with the total. A useful additional species-centered allocation is

$$\frac{s_2^{(a)}}{k_B} = -2\pi\rho \sum_{b} x_b I_{ab}, \qquad s_2 = \sum_a x_a s_2^{(a)}. \tag{5}$$

These are structural allocations, not partial molar entropy derivatives. Report $+2$ and -2 separately and their equal-population average.

Define selected probe metrics carefully. The weighted central-probe allocation $\sum_{a\in\{+2,-2\}} x_a s_2^{(a)}$ counts a probe–neutral term once in the ordered sum, whereas the union of all unordered pairs involving a probe contains both ordered directions of that cross term. They are distinct observables. Report both with explicit definitions rather than calling both local excess entropy. Compare them to the full sum and to the He et al. selected-pair construction only after resolving its normalization.

5 Bias, radial convergence, and mandatory validation

The function $\phi(g)$ is convex. Histogram noise therefore creates a negative bias in s_2, particularly in sparsely populated ionic bins and nominally flat tails. A small statistical error bar does not eliminate this systematic bias. Use a pooled RDF within each replica for the primary estimate: the entropy of the pooled RDF and the mean of entropies of small-block RDFs are different estimators. Blocks diagnose uncertainty and sample-size bias; the latter mean is not substituted silently for the former.

Required checks are:

1. Independent ideal random-point samples, with the same box, counts, frame counts, binning, and normalization. The expectation of the RDF is one, but the plug-in entropy is generally negatively biased at finite sample size. Quantify this bias instead of assuming the observed ideal-gas entropy is zero. Do not subtract its measured bias from interacting data without validation.

2. A small-frame independent direct-distance/histogram cross-check; translation, atom-order and type-code invariance; RDF symmetry; self-pair exclusion; exact pair-count bookkeeping; and recovery of the single-species formula when physical species are merged.

3. Nonpositivity of every converged pair contribution, correct hard-core integration, total/species allocation additivity, and the exact synthetic label split/merge null. Statistical differences of label-resolved plug-in estimates must not be interpreted as physical entropy changes.

4. Sample-count and bin-width convergence, raw rare-bin counts, first/second half stationarity, and effective correlation times. More closely spaced frames are not automatically independent configurations.

5. Cumulative entropy curves $s_2(R)$ and pair-resolved tails. Compare $R = 4, 5, 6, 7, 8\sigma$ and the allowed maximum, with particular attention to ionic pairs and the small controlled increments. Do not set $g = 1$ beyond a chosen radius merely to force a plateau.

An optional smooth or fitted tail must be fixed independently of TI or diffusion, validated on held-out radii, and reported separately from the directly sampled integral. HNC can guide a tail sensitivity but cannot serve as independent MD confirmation. If no radial plateau is resolved within the box, report a truncated $s_2(R)$ with a systematic limitation and plan a larger box. Also compare plausible finite-system RDF normalization conventions; an ideal-normalized RDF does not remove all finite-size effects in correlations.

6 Statistical analysis and comparison with TI and diffusion

Produce one converged estimate per independent replica. Use paired composition contrasts across the three matched replica indices, with nominal 95% Student-t intervals and two degrees of freedom, consistent with the current study. Within-replica block resampling or a validated block jackknife can quantify histogram uncertainty and bias, but cannot create additional MD replicas. Correlated frame or pair counts are not the sample size for the between-replica confidence interval. Retain covariance when comparing TI and s_2 from the same replica.

The main thermodynamic outputs are s_2^{MD}, $s_{\mathrm{ex}}^{\mathrm{TI}}$, $r_{\geq 3} = s_{\mathrm{ex}}^{\mathrm{TI}} - s_2^{\mathrm{MD}}$, and their paired increments. Interpret $r_{\geq 3}$ as the residual beyond the pair approximation under matched conventions; unresolved RDF, TI, finite-size, and tail errors also enter it. Evaluate s_2/s_{ex} only with uncertainty and a common normalization. Do not impose a 90% pair fraction. Compare HNC and MD RDFs and s_2 at the same density, temperature, charges, composition, and potential convention; verify the input LJ shift/cutoff and numerical radial conventions. Do not mix HNC s_2 with MD total entropy when estimating the MD residual.

For dynamics retain the existing primary 20–100τ MSD fitting window, with 10–50 and 50–200τ checks. Compare $\ln D^{\mathrm{R}}$ with full TI entropy, full MD s_2, and the predefined probe metrics. Use $D_i^{\mathrm{R}} = D_i \rho^{1/3} \sqrt{m_i/(k_B T)}$, with total number density. Assess mean self-diffusion $\sum_i x_i D_i$, each divalent sign, and the pooled probe mean separately. These are not conductivity or mutual diffusion.

Exclude L5 from regression counts; fit fixed-ionic-strength composition families separately. With only seven physical states, restrict initial models to one entropy coordinate plus an intercept, show residuals and paired contrasts, and use leave-one-state-out or cross-background prediction. Uncertainty in both entropy and diffusion and sensitivity to individual states must be shown. A high R^2 from a few composition means is not evidence of a universal law. No RDF smoothing, radius, or structural definition is selected to maximize agreement with diffusion.

7 Execution stages, decisions, and deliverables

Stage	Work and decision
A: inventory/pilot	Verify all 24 files, IDs, physical species, time windows, and boxes. Benchmark B3 and H7, establish a tested histogram kernel, and pass ideal-gas and synthetic-label checks before full processing.

Stage	Work and decision
B: all-pair RDFs	Process all eight states and three replicas. Increase frame counts and vary bins/radii until the selected contrasts are stable or a sampling limitation is demonstrated. Preserve histograms for reanalysis.
C: entropy benchmark	Compare MD s_2 and selected contributions with TI and HNC. Proposed practical targets are a 95% half-width below $0.005k_B$ for controlled entropy increments and bin/frame/radial sensitivity below $0.002k_B$. These are design goals, not guarantees or a formal equivalence test.
D: transport assessment	Test whether entropy ordering, controlled contrasts, and descriptive diffusion relationships survive replacement of TI by pair entropy. Changing an entropy estimator does not resolve the existing uncertain 5% probe diffusion response.
E: conditional new MD	If sampling is limiting, extend dense records or add replicas. If radial/finite-size errors dominate, compare a second size, for example $N = 8000$ at the same state and fractions, using the same physical LJ potential and controlled electrostatic accuracy. Submit only after a separate execution decision; no new jobs are required for the initial extraction.

The dense trajectories suffice for the primary entropy–diffusion question. Positions at all 19 TI densities are not established as available: the production writer records all-particle position trajectories at the dense transport stage. A density-dependent $s_2(\rho)$ extension therefore requires an inventory of any additional stored configurations or new sampling. Prioritize a few representative densities for B3, U4, H5, U6, and H7 only if the dense benchmark indicates a scientific need. Higher concentration and size/charge asymmetry are later transfer studies, not prerequisites for the current all-pair calculation.

Write a separate analysis directory, for example `MDPairEntropyAssessment/`, containing a protocol/configuration file, source and trajectory checksums, software versions, per-replica raw pair counts and RDFs, cumulative pair integrals, full and selected s_2 values, noise/tail/bin/frame checks, paired contrast intervals, TI residuals, and transport-comparison plots. Preserve machine-readable CSV/JSON summaries and the exact frame manifest. The final report should explicitly decide whether pair entropy changes the qualitative conclusions, quantitative increments, or species-versus-bulk interpretation, and identify any questions that remain unresolved rather than forcing agreement with the HNC expectation.

References

[1] Y. Rosenfeld, *Phys. Rev. A* **15**, 2545–2549 (1977). doi:10.1103/PhysRevA.15.2545.

[2] W. P. Krekelberg et al., "Generalized Rosenfeld scalings for tracer diffusivities in not-so-simple fluids: Mixtures and soft particles," *Phys. Rev. E* **80**, 061205 (2009). doi:10.1103/PhysRevE.80.061205.

[3] L. He et al., "Bridging thermodynamics and ion transport in lithium batteries via molecular design of high-entropy electrolytes," *Molecular Systems Design & Engineering* **11**, 681–690 (2026). doi:10.1039/D6ME00060F. See the full-text assessment in `doc/entropy_diffusion_prior_work.tex`.